

Ontwikkelperslag G3-recombined

Opgemaakt 2026-10-09 12:16 CEST uit het generatieregister en het evolutieregister. Status:

survived

1. Selectie

Besluit vastgelegd in het evolutieregister (rij 41). Onderbouwing:

```
{
  "aggregate_iqm": 0.9449376861365649,
  "approval_sha256": "39a0c6258c4c16e026c1382a35bbb494df9017129bfb68116e94f3369aae0a5b",
  "approved_at": "2026-10-09T10:14:05.947908+00:00",
  "limitations": [
    "UnlockPickup IQM 0 on both secret sets",
    "about 5x G2 weight storage; serving energy and latency unmeasured"
  ],
  "promotion_vs_G2": {
    "candidate_iqm": 0.9457662001266569,
    "p_value": 0.003968253968253968,
    "prob_improvement": 1.0,
    "promote": true,
    "reference_iqm": 0.8115200559194508
  },
  "promotion_vs_random_control": {
    "candidate_iqm": 0.9457662001266569,
    "p_value": 0.015873015873015872,
    "prob_improvement": 0.92,
    "promote": true,
    "reference_iqm": 0.8025422676783659
  },
  "replication_report_sha256": "429064212fa6258d5956ffa2064e252a510623b76fd470219893b68cc88bbe0a",
  "review_sha256": "99b98ac8238654fc9cb3621eeel592602d1fc91575b5c6e479ce5b9dd2a62755",
  "reviewed_by": "knight2",
  "signature_sha256": "9e79a8d38f4c35c659490282639697f29c642cb6f3376f76938b0a85de2fce2d",
  "trial_report_sha256": "d3ba19a2b52f343f52b67b8426f33c350e4d580ae99357b9897f0e2c038dace4"
}
```

2. Populatie

toddler	rol	registratie	geërfd van (modules)
G3-recombined/ t5001	population	born	G2/t2001: task:cartpole, task:acrobot, task:empty5, task:doorkey5, task:doorkey8, task:unlock G2-scratch/t2001: G3-sp/t3001: task:unlockpickup, task:keycorridor3 G3-trunk/t3001: G3-scratch/t3001: task:lavacross9
G3-recombined/ t5002	population	born	G2/t2002: task:cartpole, task:acrobot, task:empty5, task:doorkey5, task:unlock, task:unlockpickup, task:keycorridor3 G2-scratch/t2002: task:doorkey8, task:lavacross9 G3-sp/t3002: G3-trunk/t3002: G3-scratch/t3002:
G3-recombined/ t5003	population	born	G2/t2003: task:cartpole, task:acrobot, task:empty5, task:doorkey5, task:unlock, task:unlockpickup G2-scratch/t2003:

		G3-sp/t3003: G3-trunk/t3003: task:doorkey8, task:keycorridor3 G3-scratch/t3003: task:lavacross9
G3-recombined/ t5004	population born	G2/t2004: task:cartpole, task:acrobot, task:empty5, task:doorkey5, task:unlock, task:unlockpickup, task:keycorridor3, task:lavacross9 G2-scratch/t2004: G3-sp/t3004: G3-trunk/t3004: G3-scratch/t3004: task:doorkey8
G3-recombined/ t5005	population born	G2/t2005: task:cartpole, task:acrobot, task:empty5, task:doorkey5, task:unlock, task:unlockpickup G2-scratch/t2005: task:doorkey8, task:keycorridor3, task:lavacross9 G3-sp/t3005: G3-trunk/t3005: G3-scratch/t3005:

3. Overerving (generatiedata)

kind	ouder	sha256 oudergewichten
G3-recombined/t5001	G2/t2001	129643dcbfbf35ac752eeb3bbf0a3858a08d107ca1914a353510265d2c5eec1f
G3-recombined/t5001	G2-scratch/t2001	b593b7395016ea78a1e8ebb643ec6a613d72829d5d6948758845b42b0f20583e
G3-recombined/t5001	G3-sp/t3001	49bb5103d45d5a94f6240d0ff533d86f253d4b890732f8beba6a84860bdc4b92
G3-recombined/t5001	G3-trunk/t3001	ca5eb55a2a655da37ab4ab22d9f0e5ee4a46968709c3d8db82b1f46e6e4f20ed
G3-recombined/t5001	G3-scratch/t3001	25aede4f5ce87f3973e7827243af173bbe40ddd5d1fdb7ba7880657fc50c9ae
G3-recombined/t5002	G2/t2002	6aba323367f8a0967f9028e1c51fabef584808ca08f31f0f04685b915c5accca
G3-recombined/t5002	G2-scratch/t2002	cd7c85c7bc2aabf632d65a9a1a623df979355d5b065fd28b7ace2e51bb383112
G3-recombined/t5002	G3-sp/t3002	c40e3683f452f49e2e50e54394c08c897a32d90f9d82dc7ca37e6293b8224138
G3-recombined/t5002	G3-trunk/t3002	a05f776220ebddd4c3fa70dcd7cc394858fa3620505630c6d5c7be1cbb329fb5
G3-recombined/t5002	G3-scratch/t3002	3333fc18baa3a7e5df7560395aae09e73d774ac1034ce10c2da37223717fec03
G3-recombined/t5003	G2/t2003	16ec22df7a650c0b720a7f5125b794213ffed73aa1cba6616d193a679a6a6087
G3-recombined/t5003	G2-scratch/t2003	812a2b5dc2d002d9c2df6b24c3b04ca1fef807c413dbaba5c28351ce9a4b0d8b
G3-recombined/t5003	G3-sp/t3003	988afc781d2a63d08d30f1c948eb9e0ac83b41116e989d4fceed0ef1cc21b411
G3-recombined/t5003	G3-trunk/t3003	a915a6d1f6f8bd8be78fffad65020b31826a03879e007f687f417728b7036398
G3-recombined/t5003	G3-scratch/t3003	7bfe21c4e047d91e96a1d811f8821111e6549a32e7d542df91a97bbc037e4def
G3-recombined/t5004	G2/t2004	dfde4eec4229e5e954567e8daecbb5830a18d5e8fe71d9022b001767d05d327b
G3-recombined/t5004	G2-scratch/t2004	ae5281e60243c04db4ee3e973e52e2a29aefde73e42f26b2cb0760b9532986d4
G3-recombined/t5004	G3-sp/t3004	cdcf1d25b16bae03c21ee3e8659476cc227a31dbe05fa6f855e5aecb304fbd0f0
G3-recombined/t5004	G3-trunk/t3004	4ea321eb3b6be829cb7bb4585a03fa6f4d61ea025400fdd656b5ee6a686dc7cc
G3-recombined/t5004	G3-scratch/t3004	3f2be853243b2233e78c5b302c8ca17506663529450f5c34c8cd96be50fca604
G3-recombined/t5005	G2/t2005	811164b29af7821321e44c561d15063ac5855e5d2766ad47d3bec902b76521cc
G3-recombined/t5005	G2-scratch/t2005	31f472cd25a74f9724c536acd9b449324673af25503b3905d8fbdf523a2e358
G3-recombined/t5005	G3-sp/t3005	4b9cbefb1aaaf85c7508033bd89180d4cd16bd0cf278ed5e2e199ee33004bd5

G3-recombined/t5005	G3-trunk/t3005	cdd3f2f8142bc305b1176dc636ef286a2ec0b94007a7979f66cea43796fd2101
G3-recombined/t5005	G3-scratch/t3005	0d4d2576754ce86c6b4e5980744879ea8f0867abf6326aee24a16eb15adcbd5c

Elke geërfde ouder is bij de geboorte gecontroleerd tegen zijn eigen registratie; een afwijkende hash wordt door het register geweigerd.

4. Stamboom

G1	G2	G3-sp	G3-trunk	G2
t1001	t2001	t3001	t3001	t2001
t1002	t2002	t3002	t3002	t2002
t1003	t2003	t3003	t3003	t2003
t1004	t2004	t3004	t3004	t2004
t1005	t2005	t3005	t3005	t2005

5. Hardware

```
{
  "cores": 96,
  "cpu": "x86_64",
  "device": "cpu",
  "gpus": [
    "NVIDIA RTX 4000 Ada Generation",
    "NVIDIA RTX 4000 Ada Generation",
    "NVIDIA RTX 4000 Ada Generation",
    "Tesla V100-SXM2-32GB",
    "Tesla V100-SXM2-32GB",
    "NVIDIA RTX 4000 Ada Generation"
  ],
  "torch_threads": 48
}
```

6. Software en code

```
{
  "commit": "9b505a4a67873dff8c83621b7b86dc57d1308b8a",
  "dirty": false,
  "remote": "git@github.com:virtuanalytica/toddler.git"
}

{
  "gymnasium": "1.2.1",
  "numpy": "2.3.1",
  "torch": "2.6.0+cu124"
}
```

7. Trainingsdata en evaluatie

```
{
  "protocol_sha256": "fbdb806fd77ee14e6519ab3efb774a484417a07649a38f3f95c59059074e359d",
  "replication_protocol_sha256": "b7b7d8277da2236fe1516f7e6f47a4cc1964267d4777c1ebea543334342bc8ca",
}
```

```
"review_sha256": "99b98ac8238654fc9cb3621eee1592602d1fc91575b5c6e479ce5b9dd2a62755"
}
```

Een verborgen seed-set wordt alleen met zijn commitment vermeld tot het venster sluit; daarna is hij te controleren met `python3 -m toddler.learn.secret_seeds reveal <set_id>`.

8. Budget

toddler	budget
G3-recombined/t5001	{"method": "frozen-task-expert-route", "new_environment_steps": 0}
G3-recombined/t5002	{"method": "frozen-task-expert-route", "new_environment_steps": 0}
G3-recombined/t5003	{"method": "frozen-task-expert-route", "new_environment_steps": 0}
G3-recombined/t5004	{"method": "frozen-task-expert-route", "new_environment_steps": 0}
G3-recombined/t5005	{"method": "frozen-task-expert-route", "new_environment_steps": 0}

9. Resultaten (generatierapport)

```
{
  "protocol_sha256": "fbdb806fd77ee14e6519ab3efb774a484417a07649a38f3f95c59059074e359d",
  "secret_seed_commitment": {
    "set_id": "20261008T175942Z-5b684c",
    "n": 30,
    "seed_low": 100000,
    "seed_high": 1000000,
    "created_at": "2026-10-08T17:59:42+00:00",
    "reveal_after": "2026-10-15",
    "commitment": "e756a6526f000ef7e207e3e974c3388991770e35f53da32c2443db3cc7dda7a8"
  },
  "selection_rule": "On fresh secret seeds, candidate must pass decide_promotion versus frozen G2 and an equal-expert-bank",
  "routes": {
    "t2001": {
      "cartpole": "G2",
      "acrobot": "G2",
      "empty5": "G2",
      "doorkey5": "G2",
      "doorkey8": "G2",
      "unlock": "G2",
      "unlockpickup": "G3-sp",
      "keycorridor3": "G3-sp",
      "lavacross9": "G3-scratch"
    },
    "t2002": {
      "cartpole": "G2",
      "acrobot": "G2",
      "empty5": "G2",
      "doorkey5": "G2",
      "doorkey8": "G2-scratch",
      "unlock": "G2",
      "unlockpickup": "G2",
      "keycorridor3": "G2",
      "lavacross9": "G2-scratch"
    },
    "t2003": {
      "cartpole": "G2",
      "acrobot": "G2",
      "empty5": "G2",
      "doorkey5": "G2",
      "doorkey8": "G3-trunk",
      "unlock": "G2",
      "unlockpickup": "G2",
    }
  }
}
```

```

    "keycorridor3": "G3-trunk",
    "lavacross9": "G3-scratch"
  },
  "t2004": {
    "cartpole": "G2",
    "acrobot": "G2",
    "empty5": "G2",
    "doorkey5": "G2",
    "doorkey8": "G3-scratch",
    "unlock": "G2",
    "unlockpickup": "G2",
    "keycorridor3": "G2",
    "lavacross9": "G2"
  },
  "t2005": {
    "cartpole": "G2",
    "acrobot": "G2",
    "empty5": "G2",
    "doorkey5": "G2",
    "doorkey8": "G2-scratch",
    "unlock": "G2",
    "unlockpickup": "G2",
    "keycorridor3": "G2-scratch",
    "lavacross9": "G2-scratch"
  }
},
"secret": {
  "candidate": {
    "aggregate_iqm": 0.9449376861365649,
    "per_child_mean": [
      1.0582790969624787,
      0.9354701969856417,
      0.9462893364646061,
      0.947477514097081,
      0.9435317498182835
    ],
    "per_task_iqm": {
      "cartpole": 1.0553342186808325,
      "acrobot": 1.0454444444444444,
      "empty5": 1.0832450431360674,
      "doorkey5": 1.0704362698031689,
      "doorkey8": 1.082843795620438,
      "unlock": 1.074922839506173,
      "unlockpickup": 0.0,
      "keycorridor3": 1.0581590660955742,
      "lavacross9": 1.0340534979423868
    }
  },
  "random_control": {
    "aggregate_iqm": 0.8432622311136746,
    "per_child_mean": [
      0.94523363700895,
      0.864782873702419,
      0.8190497214988377,
      0.7086777125157392,
      0.7237942078338407
    ],
    "per_task_iqm": {
      "cartpole": 1.0553342186808325,
      "acrobot": 1.0454444444444444,
      "empty5": 1.0832450431360674,
      "doorkey5": 1.0704362698031689,
      "doorkey8": 0.41003990267639906,
      "unlock": 1.0744598765432098,
      "unlockpickup": 0.0,
      "keycorridor3": 1.0515663055345597,
      "lavacross9": 0.7988340192043895
    }
  }
}

```

```

},
"G2": {
  "aggregate_iqm": 0.771915604564624,
  "per_child_mean": [
    0.824933399286975,
    0.7978242719942359,
    0.8118024964771416,
    0.8260273924425796,
    0.6492463894378111
  ],
  "per_task_iqm": {
    "cartpole": 1.0553342186808325,
    "acrobot": 1.0454444444444444,
    "empty5": 1.0832450431360674,
    "doorkey5": 1.0704362698031689,
    "doorkey8": 0.022909975669099764,
    "unlock": 1.074922839506173,
    "unlockpickup": 0.0,
    "keycorridor3": 0.6709078693205678,
    "lavacross9": 0.924039780521262
  }
},
"G2-scratch": {
  "aggregate_iqm": 0.8826871472322377,
  "per_child_mean": [
    0.8881624521998143,
    0.9352261624418738,
    0.8006363299789654,
    0.7620543326682512,
    0.9412453427585861
  ],
  "per_task_iqm": {
    "cartpole": 1.05317003590576,
    "acrobot": 1.0368611111111111,
    "empty5": 1.0832450431360674,
    "doorkey5": 1.0674657945271244,
    "doorkey8": 0.7118189781021899,
    "unlock": 1.0701003086419754,
    "unlockpickup": 0.0,
    "keycorridor3": 1.05660535819266,
    "lavacross9": 0.864917695473251
  }
},
"G3-sp": {
  "aggregate_iqm": 0.8463294866630688,
  "per_child_mean": [
    0.912765164627403,
    0.7078367275090021,
    0.8053950710360045,
    0.9475912830628059,
    0.8102439770808645
  ],
  "per_task_iqm": {
    "cartpole": 1.0553342186808325,
    "acrobot": 1.0439444444444443,
    "empty5": 1.0827909792644161,
    "doorkey5": 1.0687259961593856,
    "doorkey8": 0.40098102189781026,
    "unlock": 1.0766203703703703,
    "unlockpickup": 0.0,
    "keycorridor3": 1.057361216091375,
    "lavacross9": 0.8312071330589849
  }
},
"G3-trunk": {
  "aggregate_iqm": 0.8055301569899499,
  "per_child_mean": [
    0.8329403050780583,

```

```

0.7475914958202533,
0.8469222749638469,
0.8959266023652592,
0.7286957106471192
],
"per_task_iqm": {
  "cartpole": 1.0553342186808325,
  "acrobot": 1.0378055555555556,
  "empty5": 1.0832450431360674,
  "doorkey5": 1.0676908305328852,
  "doorkey8": 0.4646209245742092,
  "unlock": 1.0744984567901235,
  "unlockpickup": 0.0,
  "keycorridor3": 1.0561854371378183,
  "lavacross9": 0.41039094650205765
}
},
"G3-scratch": {
  "aggregate_iqm": 0.8591724894863382,
  "per_child_mean": [
    0.659223927727886,
    0.814624118749072,
    0.9456687235093961,
    0.8246684899907003,
    0.8192987457660684
  ],
  "per_task_iqm": {
    "cartpole": 1.0553342186808325,
    "acrobot": 1.0361666666666667,
    "empty5": 1.082185560768881,
    "doorkey5": 1.0633701392222756,
    "doorkey8": 0.3552642335766423,
    "unlock": 1.0742283950617284,
    "unlockpickup": 0.0,
    "keycorridor3": 1.057361216091375,
    "lavacross9": 1.008641975308642
  }
}
},
"promotion_vs_G2": {
  "promote": true,
  "p_value": 0.003968253968253968,
  "prob_improvement": 1.0,
  "candidate_iqm": 0.9457662001266569,
  "reference_iqm": 0.8115200559194508
},
"promotion_vs_random_control": {
  "promote": true,
  "p_value": 0.015873015873015872,
  "prob_improvement": 0.92,
  "candidate_iqm": 0.9457662001266569,
  "reference_iqm": 0.8025422676783659
},
"beats_all_source_IQMs": true,
"preserves_G1_tasks": true,
"eligible_for_lineage_review": true,
"verdict": "eligible successor for review",
"children": [
  {
    "arm": "candidate",
    "id": "t5001",
    "weights_sha256": "143b97dbdd92496b8d7187366ed32da9b21f59103b67eea6466a3cf7e681e0d6"
  },
  {
    "arm": "random_control",
    "id": "t5001",
    "weights_sha256": "9003ddd190a681cc383313b577f0eedc66e1d6a82b931bf3fa00dc5a4df41e71"
  }
],

```

```

{
  "arm": "candidate",
  "id": "t5002",
  "weights_sha256": "b78134381192b713e5a67b8748dea3eb7ea603a2636e60042d9b85f48abb9c15"
},
{
  "arm": "random_control",
  "id": "t5002",
  "weights_sha256": "38ce371b88405463dafd3356c38fcf577b26ff74db76d77364238219fb10006e"
},
{
  "arm": "candidate",
  "id": "t5003",
  "weights_sha256": "855c89a66c3414e5470b95fc0b1e3ef97ba43af7c10be2a14e79551b46a054be"
},
{
  "arm": "random_control",
  "id": "t5003",
  "weights_sha256": "45a56168b7ffe5ff54e4d9401bf4d8fff1aad7d03b4ca070f75747334c5c72bd"
},
{
  "arm": "candidate",
  "id": "t5004",
  "weights_sha256": "17ae7be6e8ecca01057ad6dd7f9425ab819a4e41f10c70dae6620fb30937470f"
},
{
  "arm": "random_control",
  "id": "t5004",
  "weights_sha256": "03b615cfaf9745d33af34a709bcdaab857861f36c7a67873667c33a8d00e5972"
},
{
  "arm": "candidate",
  "id": "t5005",
  "weights_sha256": "95af37f23ca7ded3e78f23a7e1c966bb5d798866c75cc888d6bab75fda0672f2"
},
{
  "arm": "random_control",
  "id": "t5005",
  "weights_sha256": "115d3f14f74926190da21ecdc2dbcce67e5409ad6337237b2b91eb3530d1e3f8"
}
],
"duration_s": 89.3
}

```

10. Afgeleide agents en transmorphoses

agent	soort	doel	goedgekeurd door
-	-	nog geen agents op deze generatie gebouwd	-

11. Integriteit

toddler	sha256 gewichten (nu gemeten)	register + meta.json
G3-recombined/ t5001	143b97dbdd92496b8d7187366ed32da9b21f59103b67eea6466a3cf7e681e0d6	ok
G3-recombined/ t5002	b78134381192b713e5a67b8748dea3eb7ea603a2636e60042d9b85f48abb9c15	ok
G3-recombined/ t5003	855c89a66c3414e5470b95fc0b1e3ef97ba43af7c10be2a14e79551b46a054be	ok

G3-recombined/ t5004	17ae7be6e8ecca01057ad6dd7f9425ab819a4e41f10c70dae6620fb3093747 0f	ok
G3-recombined/ t5005	95af37f23ca7ded3e78f23a7e1c966bb5d798866c75cc888d6bab75fda0672 f2	ok

Hashketen van het evolutieregister: intact; 42 rijen; laatste rijhash d2496dea439a9371b8bf290308e9021ea1cd01998449b5dce1f2b917db64073d.